

Practical 03

Question

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Process started\n");

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
    }
    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        printf("Child is running\n");

        sleep(5);

        printf("Child process terminated\n");
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Parent is running\n");

        sleep(10);

        printf("Parent process terminated\n");
    }

    return 0;
}
```

```

bhavan@LAPTOP-FGDPB110:~$ nano week3_task3.c
bhavan@LAPTOP-FGDPB110:~$ gcc week3_task3.c -o week3_task3
bhavan@LAPTOP-FGDPB110:~$ ./week3_task3
Process started
Parent Process
Parent PID: 831
Parent is running
Child Process
Child PID: 832
Parent PID: 831
Child is running
Child process terminated
Parent process terminated
bhavan@LAPTOP-FGDPB110:~$ ps
  PID TTY          TIME CMD
  705 pts/0    00:00:00 bash
  833 pts/0    00:00:00 ps
bhavan@LAPTOP-FGDPB110:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1        0  0   13:31 ?        00:00:01 /sbin/init
root           2        1  0   13:31 ?        00:00:00 /init
root           6        2  0   13:31 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          58        1  0   13:31 ?        00:00:00 /usr/lib/systemd/systemd-journald
root         109        1  0   13:31 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+     123        1  0   13:31 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     124        1  0   13:31 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root         187        1  0   13:31 ?        00:00:00 /usr/sbin/cron -f -P
message+     188        1  0   13:31 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-on
root         202        1  0   13:31 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         226        1  0   13:31 tty1    00:00:00 /sbin/getty -o -p -- \u --noclear - linux
syslog       227        1  0   13:31 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         234        1  0   13:31 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         340        2  0   13:32 pts/1   00:00:00 /bin/login -f
bhavan       390       705  0   13:32 ?        00:00:00 /usr/lib/systemd/systemd --user
bhavan       391       390  0   13:32 ?        00:00:00 (sd-pam)
bhavan       415       340  0   13:32 pts/1   00:00:00 -bash
root         700        2  0   13:50 ?        00:00:00 /init
root         702       700  0   13:50 ?        00:00:00 /init
root         705       702  0   13:50 pts/0    00:00:00 -bash
root         793        1  0   14:03 ?        00:00:00 /usr/libexec/wsl-pro-service
root         834       109  0   14:06 ?        00:00:00 (udev-worker)
bhavan@LAPTOP-FGDPB110:~$ |

```

```

bhavan@LAPTOP-FGDPB110:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root         700        2  0   13:50 ?        00:00:00 /init
root         702       700  0   13:50 ?        00:00:00 /init
bhavan       705       702  0   13:50 pts/0    00:00:00 -bash
root         793        1  0   14:03 ?        00:00:00 /usr/libexec/wsl-pro-service
root         834       109  0   14:06 ?        00:00:00 (udev-worker)
root         835       109  0   14:06 ?        00:00:00 (udev-worker)
bhavan       836       705  0   14:06 pts/0    00:00:00 ps -ef
bhavan@LAPTOP-FGDPB110:~$ ps -ef | top
top - 14:07:31 up 35 min,  1 user,  load average: 0.02, 0.04, 0.00
Tasks: 22 total,  1 running, 21 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7754.6 total, 7142.6 free,  532.0 used,  230.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used,  7222.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
837	bhavan	20	0	9300	5700	3512	R	0.3	0.1	0:00.05	top
1	root	20	0	21860	13148	9800	S	0.2	0.2	0:01.69	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.03	init-systemd(ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
58	root	19	-1	50368	16184	15100	S	0.0	0.2	0:00.49	systemd-journal
109	root	20	0	25284	6712	5176	S	0.0	0.1	0:01.18	systemd-udevd
123	systemd+	20	0	21344	13244	11120	S	0.0	0.2	0:00.28	systemd-resolve
124	systemd+	20	0	91036	8000	7032	S	0.0	0.1	0:00.37	systemd-timesyn
187	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.02	cron
188	message+	20	0	9548	5344	4640	S	0.0	0.1	0:00.17	dbus-daemon
202	root	20	0	17980	8808	7764	S	0.0	0.1	0:00.17	systemd-logind
226	root	20	0	3124	1968	1836	S	0.0	0.0	0:00.01	agetty
227	syslog	20	0	222516	5964	4568	S	0.0	0.1	0:00.19	rsyslogd
234	root	20	0	107036	23256	13804	S	0.0	0.3	0:00.32	unattended-upgr
340	root	20	0	6704	4656	3872	S	0.0	0.1	0:00.03	login
390	bhavan	20	0	20324	11504	9376	S	0.0	0.1	0:00.26	systemd
391	bhavan	20	0	21168	3596	1868	S	0.0	0.0	0:00.06	(sd-pam)
415	bhavan	20	0	6080	5360	3696	S	0.0	0.1	0:00.05	bash
700	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
702	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.02	Relay(705)
705	bhavan	20	0	6080	5376	3656	S	0.0	0.1	0:00.09	bash
793	root	20	0	1830360	14308	11768	S	0.0	0.2	0:00.33	wsl-pro-service

```

bhavan@LAPTOP-FGDPB110:~$ |

```

The `fork()` system call is used to create a new process. The original process is called the **parent process**, and the newly created process is called the **child process**. Each process has its own **PID**, and the child process can get its parent's ID using **PPID**. During execution, processes move through different states such as running, waiting, and terminated.